



锂离子电池电极颗粒中的嵌入引起的应力数值模拟

Xiangchun Zhang,^{a,*} Wei Shyy,^b and Ann Marie Sastry^{a,c,d,**,z}

^a 密歇根大学机械工程, ^b 航空航天工程, ^c 生物医学工程, ^d 材料科学与工程, 美国密歇根州安阿伯市 48109-2125

在生产和循环过程中产生的严重粒子级应变已被推测与锂离子电池的寿命限制性损伤相关。由于目前尚不清楚制造和嵌入引起的应力在锂电池中的贡献,这一关联对于确定这些电池的最佳材料和制造方法至关重要。必须估算全局和局部载荷,以便选择能够抵抗断裂的材料。本文选择了LiMn₂O₄系统进行研究。我们展示了一组仿真技术的结果,涵盖从球形颗粒的一维有限差分仿真,到椭球形颗粒的完全三维(3D)仿真,系统地研究了在不同形状和尺寸的颗粒中发展起来的嵌入引起的应力,其中后者的3D计算使用了商业有限元代码进行。球形颗粒的仿真表明,较大的颗粒尺寸和较大的放电电流密度会产生更大的嵌入引起的应力。椭球形颗粒的仿真表明,较大的长短比有助于减少嵌入引起的应力。总体而言,这些结果表明,合成较小尺寸和较大长短比的电极颗粒有助于减少锂离子电池在循环过程中产生的嵌入引起的应力。

© 2007 电化学学会。[DOI: 10.1149/1.2759840] 版权所有。

在生产和循环过程中产生的严重粒子级应变已被推测与锂离子电池的寿命限制性损伤相关。锂离子在阴极晶格中的嵌入和脱嵌(包括LiCoO₂、LiMn₂O₄和LiFePO₄等材料)已被假设为会导致颗粒内的断裂,这一结果已通过模型系统的实验得到了验证。例如,在LiMn₂O₄中,当Mn₂O₄被锂化为LiMn₂O₄时,已报告观察到6.5%的体积变化;对LiMn₂O₄的仿真表明,嵌入引起的应力可能超过该材料的极限强度。此外,由于生产过程中压缩引起的电池级载荷所产生的应力,已被证明会导致局部颗粒应力,这在石墨负极材料中要高得多——局部应力与全局应力的比率约为25到140。实际上,这些应力的大小超过了这些材料的已知强度,包括最常用且最有前景的阴极材料。[表I (参考文献7、8、4和9)]

由于锂离子嵌入引起的应力,及其他过程中的应力,已在先前的研究中进行了粒子级建模。Christensen和Newman估算了锂插入到碳负极和LiMn₂O₄阴极颗粒中所产生的应力。更广泛地说,由物质扩散引起的应力已在其他领域进行过研究,包括金属氧化和半导体掺杂。Prussin首次通过类比热应力的方法研究了扩散引起的应力。在这项研究中,研究了硅晶片中硼和磷掺杂过程中的应力生成。Li也采用这一方法研究了在简单几何形状的弹性介质中扩散引起的应力或化学应力。Yang研究了薄板中化学应力的演化,考虑了化学应力与扩散之间的相互作用,并遵循Prussin热应力类比的方法。

尽管这些研究工作提供了在粒子尺度上估算应力的方法,但由于采用了不同的物理假设,到目前为止,这些方法尚未应用于三维应力问题。由于目前尚不清楚制造和嵌入引起的应力在锂电池中的贡献,这一关联对于确定锂电池的最佳材料和制造方法至关重要。必须估算全局和局部载荷,以便选择能够抵抗断裂的材料。此外,局部颗粒断裂在电池容量衰减中的作用虽然有所暗示,但由于对电池中局部载荷的普遍缺乏理解,尚未进行量化研究。

因此,本研究的重点是确定阴极颗粒中的局部颗粒应力。在这里,我们选择了LiMn₂O₄系统,参考了文献14-18中的电池性能建模,文献19和20中的结构原子尺度模拟,以及文献5中的嵌入引起的应力模拟,因为LiMn₂O₄具有低成本和环境安全的特点。我们在本研究中的目标如下:

- 1.根据热应力的类比,确定扩散引起的应力,参考文献11-13中对单个颗粒的研究,并确定与锂电池中先前研究的对应关系。
- 2.通过使用有限差分方法并验证简单结果,验证单颗粒模型的数值实现。
- 3.将该模型实现为完整的有限元方案,并模拟非球形几何颗粒中由嵌入引起的应力。

* Electrochemical Society Student Member.

** Electrochemical Society Active Member.

^z E-mail: ansastry@umich.edu

仿真

应力-应变关系——对于嵌入过程,可以假设材料的晶格常数随着插入离子的体积线性变化,这会导致应力的产生。因此,可以通过类比热应力来计算嵌入引起的应力。Prussin曾经将浓度梯度处理得类似于由温度梯度在一个未受应力的物体中生成的梯度。

包括热效应在内的应力-应变关系,经典地写作弹性体的形式:

$$\epsilon_{xx} - \alpha T = \frac{1}{E} [\sigma_{xx} - \nu(\sigma_{yy} + \sigma_{zz})]$$

表I. 阴极材料在嵌入过程中的应力和应变。

材料	测量技术	应力或者应变
LiCoO ₂ 薄膜 (参考文献7)	激光束偏转	~1 GPa (应力)
LiMn ₂ O ₄ 薄膜 (参考文献7)	激光束偏转	~0.64 GPa (应力)
LiMn ₂ O ₄ (参考文献7)	中子衍射	0.027 (应力)
LiFePO ₄ (参考文献7)	X射线衍射	0.022 (应力)

$$\epsilon_{yy} - \alpha T = \frac{1}{E} [\sigma_{yy} - \nu(\sigma_{xx} + \sigma_{zz})] \quad [1b]$$

$$\epsilon_{zz} - \alpha T = \frac{1}{E} [\sigma_{zz} - \nu(\sigma_{xx} + \sigma_{yy})] \quad [1c]$$

$$\epsilon_{xy} = \frac{\sigma_{xy}}{2G} \quad \epsilon_{yz} = \frac{\sigma_{yz}}{2G} \quad \epsilon_{xz} = \frac{\sigma_{xz}}{2G} \quad [1d]$$

其中, ϵ_{ij} 是应变分量, σ_{ij} 是应力分量, E 是杨氏模量, ν 是泊松比, G 是剪切弹性模量, α 是热膨胀系数, T 是温度相对于原始值的变化。类比地,考虑到浓度梯度的应力-应变关系可以写为:

$$\epsilon_{ij} = \frac{1}{E} [(1 + \nu)\sigma_{ij} - \nu\sigma_{kk}\delta_{ij}] + \frac{\tilde{c}\Omega}{3}\delta_{ij}$$

其中, $c = c - c_0$ 是扩散物质的浓度变化,相对于原始的“无应力”值, V^- 是溶质的部分摩尔体积。方程2可以重写,以获得应力分量的表达式:

$$\sigma_{ij} = 2\mu\epsilon_{ij} + (\lambda\epsilon_{kk} - \beta\tilde{c})\delta_{ij}$$

其中, $\mu = E/2(1 + \nu)$, $\lambda = 2\nu\mu/(1 - 2\nu)$, 和 $\kappa = V^-(3\lambda + 2\mu)/3$ 。如同在弹性学中,应变张量与位移 u 相关联,如公式(21)所示。

$$\epsilon_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)$$

忽略体力的情况下，平衡方程为公式（21）：

$$\sigma_{ij,i} = 0 \quad (j = 1, 2, 3)$$

将公式（3）和（4）代入公式（5）得到位移方程（22）：

$$\mu \nabla^2 u_i + (\lambda + \mu) u_{k,ki} - \beta \tilde{c}_{,i} = 0 \quad (i = 1, 2, 3)$$

对于单个颗粒的情况，边界条件是颗粒表面无牵引力。该条件可以表示为公式（22）：

$$p_{nx} = \sigma_{xx}l + \sigma_{yx}m + \sigma_{zx}n = 0$$

$$p_{ny} = \sigma_{xy}l + \sigma_{yy}m + \sigma_{zy}n = 0$$

$$p_{nz} = \sigma_{xz}l + \sigma_{yz}m + \sigma_{zz}n = 0$$

其中 l, m, n, l, m, n 表示外法线与每个轴之间的方向余弦。将公式（3）和（4）代入边界条件公式（7）得到：

$$\mu(u_{i,j} + u_{j,i})n_j + (\lambda u_{k,k} - \beta c)n_i = 0 \quad i = 1, 2, 3$$

其中， $n_1=l, n_2=m, n_3=n$ 。因此，我们要解的是公式（6），边界条件为公式（8）。

扩散方程——如公式（2）和（3）所示，计算嵌入引起的应力需要浓度。为了获得浓度分布，离子插入和提取被建模为扩散过程。可以忽略固体中已有电子对锂物种通量的影响，因为电子的移动性比嵌入原子要大得多（参考文献23）。化学势梯度是锂离子运动的驱动力。锂离子的速度可以表示为：

$$\mathbf{v} = -M \nabla \mu$$

其中， M 是锂离子的迁移率， μ 是化学势。然后，物种通量可以写为（参考文献23）：

$$\mathbf{J} = c\mathbf{v} = -Mc \nabla \mu$$

其中， c 是扩散组分（锂离子）的浓度。

理想固溶体中的电化学势可以表示为（参考文献13, 24）：

$$\mu = \mu_0 + RT \ln X - \Omega \sigma_h$$

这里 μ_0 是一个常数， R 是气体常数， T 是绝对温度， X 是锂离子的摩尔分数， Ω 是锂离子的部分摩尔体积， σ_h 是流体静力应力，定义为：

$$\sigma_h = \frac{\sigma_{11} + \sigma_{22} + \sigma_{33}}{3} \quad (\text{其中 } \sigma_{ij} \text{ 是应力张量中的分量})$$

方程（10）和（11）表明，扩散通量依赖于浓度、温度和应力场。将方程（11）代入方程（10），假设温度均匀，并注意到：

$$\nabla(RT \ln X) = RT \frac{1}{X} \nabla X = RT \frac{1}{c} \nabla c$$

物种通量（当粒子内部没有温度梯度时）的表达式可以得到为：

$$\mathbf{J} = -D \left(\nabla c - \frac{\Omega c}{RT} \nabla \sigma_h \right)$$

其中， $D = MRT$ 为扩散系数。物种的守恒给出：

$$\frac{\partial c}{\partial t} + \nabla \cdot \mathbf{J} = 0$$

然后，将方程（13）代入方程（14）得到：

$$\frac{\partial c}{\partial t} = D \left(\nabla^2 c - \frac{\Omega}{RT} \nabla c \cdot \nabla \sigma_h - \frac{\Omega c}{RT} \nabla^2 \sigma_h \right)$$

最终的扩散过程控制方程。初始条件为 $c=c_0$ ，边界条件为：

$$\mathbf{J} = -D \left(\nabla c - \frac{\Omega c}{RT} \nabla \sigma_h \right) = \frac{i_n}{F}$$

其中， i_n 是颗粒表面的电流密度（假设在本研究中是常数，已知值）， F 是法拉第常数。

数值方法——一维问题的有限差分法——

对于球形颗粒的情况，上述方程变为一维问题。应力张量包含两个独立分量，径向应力 σ_r 和切向应力 σ_t 。对于这种情况，平衡方程（参见方程（5））为：

$$\frac{d\sigma_r}{dr} + \frac{2}{r}(\sigma_r - \sigma_t) = 0$$

而应力-应变关系（参见方程（2））为：

$$\varepsilon_r = \frac{1}{E}(\sigma_r - 2\nu\sigma_t) + \frac{\Omega}{3}\tilde{c}$$

$$\varepsilon_t = \frac{1}{E}[\sigma_t - \nu(\sigma_r + \sigma_t)] + \frac{\Omega}{3}\tilde{c}$$

应变-位移关系（参见方程（4））为：

$$\varepsilon_r = \frac{du}{dr} \quad \varepsilon_t = \frac{u}{r}$$

而位移方程（参见方程（6））为：

$$\frac{d^2u}{dr^2} + \frac{2}{r} \frac{du}{dr} - \frac{2u}{r^2} = \frac{1+\nu}{1-\nu} \frac{\Omega}{3} \frac{d\tilde{c}}{dr}$$

对方程进行积分得到 u 的解，从中可以获得应力。注意，在球心（ $r=0$ ）处应力是有限的，而在颗粒表面（ $r=r_0$ ）处径向应力为零（ $\sigma_r=0$ ），因此可以确定解中的两个常数。

$$\sigma_r = \frac{2\Omega E}{3(1-\nu)} \left(\frac{1}{r_0^3} \int_0^{r_0} \tilde{c} r^2 dr - \frac{1}{r^3} \int_0^r \tilde{c} r^2 dr \right)$$

和

$$\sigma_t = \frac{\Omega E}{3(1-\nu)} \left(\frac{2}{r_0^3} \int_0^{r_0} \tilde{c} r^2 dr + \frac{1}{r^3} \int_0^r \tilde{c} r^2 dr - \tilde{c} \right)$$

$$\sigma_r = \frac{2KE}{3(1-\nu)} \left(\frac{1}{3} \int_0^{r_0} \tilde{c} r^2 dr - \frac{1}{3} \int_r^0 \tilde{c} r^2 dr \right) \quad [22]$$

and

$$\sigma_\theta = \frac{KE}{3(1-\nu)} \left(\frac{2}{3} \int_0^{r_0} \tilde{c} r^2 dr + \frac{1}{3} \int_r^0 \tilde{c} r^2 dr - \tilde{c} \right) \quad [23]$$

Equation 22 shows that radial stress actually depends upon the difference between the global and local averages of concentration. The diffusion equation is (referring to Eq. 15)

$$\frac{\partial c}{\partial t} = D \left[\frac{\partial^2 c}{\partial r^2} + \frac{2}{r} \frac{\partial c}{\partial r} - \frac{K}{RT} \frac{\partial c}{\partial r} + \frac{K}{RT} \left(\frac{\partial^2 c}{\partial r^2} + \frac{2}{r} \frac{\partial c}{\partial r} \right) \right] \quad [24]$$

Equations 22 and 23 allow calculation of hydrostatic stress as

$$\sigma_h = (\sigma_r + 2\sigma_\theta)/3 = \frac{2KE}{9(1-\nu)} \left(\frac{1}{3} \int_0^{r_0} \tilde{c} r^2 dr - \tilde{c} \right) \quad [25]$$

By assuming that the characteristic time for elastic deformation of solids is much smaller than that for atomic diffusion, the elastic

$$\frac{\partial c}{\partial t} = D \left[\frac{\partial^2 c}{\partial r^2} + \frac{2}{r} \frac{\partial c}{\partial r} + 0 \right] + 0c \left(\frac{\partial^2 c}{\partial r^2} + \frac{2}{r} \frac{\partial c}{\partial r} \right) \quad [26]$$

where $0 = (K/RT)[(2KE)/9(1-\nu)]$.

Substituting Eq. 25 into boundary conditions Eq. 16, one has

$$J = -D(1+0c) \frac{\partial c}{\partial r} = \frac{I_n}{F} \text{ at } r = r_0 \quad [27]$$

In this way, the two variables, concentration and stress, are decoupled.

To solve the above equation numerically, it is, along with boundary and initial condition, transformed into dimensionless form first as

$$\frac{\partial \tilde{c}}{\partial \tilde{t}} = \frac{\partial^2 \tilde{c}}{\partial \tilde{r}^2} + \frac{2}{\tilde{r}} \frac{\partial \tilde{c}}{\partial \tilde{r}} + 0 \left(\frac{\partial^2 \tilde{c}}{\partial \tilde{r}^2} + \frac{2}{\tilde{r}} \frac{\partial \tilde{c}}{\partial \tilde{r}} \right) \quad [28]$$

$$0 \leq \tilde{r} \leq 1 \quad 0 \leq \tilde{t} \leq \tilde{T} \quad [\text{where } \tilde{T} \text{ satisfies } \tilde{c}(\tilde{r}=1, \tilde{t}=\tilde{T})=1]$$

$$\tilde{r}=1 - (1+\hat{0}\tilde{c}) \frac{\partial \tilde{c}}{\partial \tilde{r}} = I \quad \tilde{r}=0 \quad \frac{\partial \tilde{c}}{\partial \tilde{r}} = 0$$

$$\tilde{t}=0 \quad \tilde{c} = c_0/c_{\max}$$

where dimensionless variables are defined as

$$\tilde{r} = \frac{r}{r_0} \quad \tilde{t} = \frac{tD}{r_0^2} \quad \tilde{c} = \frac{c}{c_{\max}} \quad \hat{0} = 0c_{\max} \quad I = \frac{i_n r_0}{Dc_{\max} F}$$

In the above equations, $c_{\max}$ is the stoichiometric maximum concentration and c_0 is the initial concentration. It may be seen that the effect of discharge current density, particle radius, and diffusion coefficient are all combined into the dimensionless current density I .

The numerical procedure is as follows. For each time step, concentration distribution is solved first by Eq. 28. Then, the concentration is substituted into Eq. 22 and 23 to calculate stresses. Equation 28 is a nonlinear, parabolic partial differential equation. The finite difference method is used here to solve the equation.

First, Eq. 28 is rewritten as

$$\frac{\partial \tilde{c}}{\partial \tilde{t}} = \frac{\partial^2 \tilde{c}}{\partial \tilde{r}^2} + \frac{2}{\tilde{r}} \frac{\partial \tilde{c}}{\partial \tilde{r}} + 0 \left(\frac{\partial^2 \tilde{c}}{\partial \tilde{r}^2} + \frac{2}{\tilde{r}} \frac{\partial \tilde{c}}{\partial \tilde{r}} \right) \quad [29]$$

To discretize the differential equation into difference equations, the problem is linearized by taking the value from the previous time

Table II. Material properties of Mn₂O₄.

Parameter	Symbol and dimensions	Value
Young's modulus	E (GPa)	10 (Ref. 25)
Poisson's ratio	ν	0.3 (Ref. 25)
Diffusion coefficient	D (m ² /s)	7.08×10^{-15} (Ref. 5)
Fatigue molar volume	V_m (m ³ /mol)	2.29×10^{-4}
Stoichiometric maximum concentration	$c_{\max}$ (mol/m ³)	2.29×10^4

step for the terms in the two parentheses on the right side. The

Crank-Nicolson method is used for other terms. The difference equation obtained is

$$\frac{c_i^{n+1} - c_i^n}{\Delta t} = D \left[\frac{c_{i+1}^{n+1} + c_{i+1}^n - 2c_i^{n+1}}{(\Delta r)^2} + \frac{c_{i+1}^n + c_{i-1}^n - 2c_i^n}{(\Delta r)^2} \right] + \frac{2}{\Delta r} \left(\frac{c_{i+1}^{n+1} - c_{i+1}^n}{\Delta t} - \frac{c_i^{n+1} - c_i^n}{\Delta t} \right) \quad [30]$$

Terms including $1/\tilde{r}$ will be singular at the particle center $\tilde{r}=0$. To solve this difficulty, noting that

$$\frac{\partial \tilde{c}}{\partial \tilde{r}} = 0 \quad \text{when } \tilde{r}=0 \quad [31]$$

L'Hopital's rule

$$\lim_{\tilde{r} \rightarrow 0} \frac{1}{\tilde{r}} \frac{\partial \tilde{c}}{\partial \tilde{r}} = \frac{\partial^2 \tilde{c}}{\partial \tilde{r}^2} \quad [32]$$

can be used to eliminate the $1/\tilde{r}$ factor. Thus, Eq. 28 becomes

$$\frac{\partial \tilde{c}}{\partial \tilde{t}} = \frac{\partial^2 \tilde{c}}{\partial \tilde{r}^2} + \frac{\partial^2 \tilde{c}}{\partial \tilde{r}^2} \quad [33]$$

which has no singularity at $\tilde{r}=0$. Then

$\tilde{r}=0$ while Eq. 29 is solved elsewhere.

At two boundary points, imaginary points (out of the boundary) are used to discretize the governing equation; the concentration values of these imaginary points are obtained by central differencing of the flux boundary condition at the boundary points.

The Thomas algorithm is used to solve the tridiagonal system of the difference equations. The simulation is halted when the concentration on the particle surface $\tilde{r}=1$ reaches unity.

Finite element method for 3D problem

simulated using FEMLAB (COMSOL Multiphysics). Two models are included in the multiphysics simulation, PDE (partial differential equation) model (general form) and solid stress-strain model. In PDE model, the diffusion process is described by the generalized

$$\frac{\partial c}{\partial t} + Q \cdot \nabla c = 0 \quad [34]$$

where

$$T = -D \left(Qc - \frac{K}{RT} \sigma_h \right) \quad [35]$$

In the solid stress-strain model, "thermal expansion" is included as a load based on the variable of concentration c instead of temperature in thermal stress calculation.

Material properties.— All the material properties used in the simulation for Mn₂O₄ are listed in Table II.^{5,25} From Eq. 2, we see that that partial molar volume plays a role analogous to a thermal

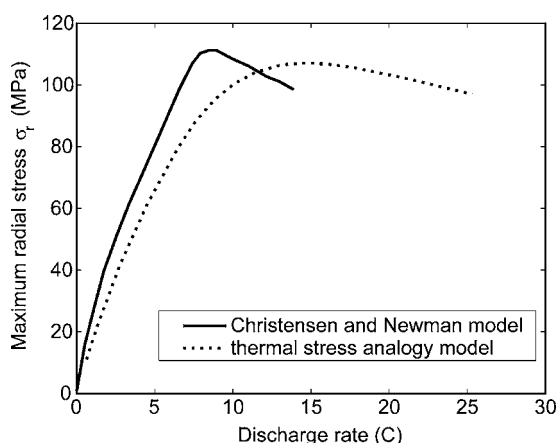


Figure 1. Comparison of simulation results of two models.

expansion coefficient, in calculating intercalation-induced stress. To obtain the value for this property, the volume change of 6.5% for $y = 0.2$ to $y = 0.995$ of $\text{Li}_y\text{Mn}_2\text{O}_4$ is used.⁵ The volume change of 6.5% gives a strain of 0.0212, which corresponds to the concentration change as $y = 0.2$ to $y = 0.995$. Therefore, partial molar volume is, by noting the analogy between thermal expansion coefficient and $K/3$

$$K = \frac{0.0212 \times 3}{(0.995 - 0.2)c_{\max}} = 3.497 \times 10^{-6} \text{ m}^3/\text{mol}$$

Results and Discussion

1D finite difference simulations.— Christensen and Newman⁵ modeled the stress generated in $\text{Li}_y\text{Mn}_2\text{O}_4$ during lithium intercalation on the 4 V plateau ($0.2 < y < 1$). The same parameters and properties are used here, except for the diffusion coefficient. In their simulation, they used a state of charge-dependent diffusion coefficient, which includes a binary interaction parameter and a thermodynamic factor. Here, a constant diffusion coefficient, taking the value of the reference binary interaction parameter in their paper, is used. The simulation results from the thermal stress analogy model and the Christensen and Newman model are shown in Fig. 1. Although different approaches are applied to calculate the intercalation-induced stress, the results qualitatively show the same trend.

We used the 1D model to simulate cycling of the active material between $y = 0$ and $y = 1$, giving an initial condition for Eq. 26 of $c_0 = 0$. Results show that that maximum radial stress is located at the center of the particle. The magnitude of the spatial maximum dimensionless radial stress is given by

$$\sigma_{r,\max} = \frac{\sigma_{r,\max}}{E} = \frac{2Kc_{\max}}{3(1-\nu)} \left(\int_0^1 \hat{c} \hat{r}^2 d\hat{r} - \frac{1}{3} \hat{c} \right)_{\hat{r}=0} \quad [36]$$

Figure 2 shows how dimensionless maximum radial stress $\sigma_{r,\max}$ (both temporally and spatially during the discharge process) varies with dimensionless current density (or dimensionless boundary flux) I .

As shown in Fig. 2, maximum radial stress (spatially and temporally) inside an electrode particle during the discharge process increases with increasing dimensionless current density when $0 < I < 2.7$. However, maximum radial stress decreases with increasing dimensionless current density when it is larger than 2.7. The decrease in stress is attributable to the fact that the concentration profile is not fully developed, so that the global average term (first term in the parentheses) in Eq. 36 decreases with dimensionless current density, while the local average (second term in the parentheses) remains constant. This is not desirable in the cycling of bat-

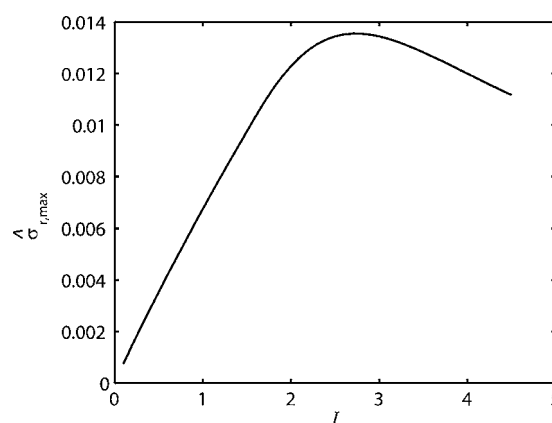


Figure 2. Maximum dimensionless radial stress vs dimensionless current density.

teries, because it reduces the utilization of the material. Therefore, only the increasing branch of the curve is actually feasible. The increasing branch shows that increase of discharge current density and particle radius will increase the intercalation-induced stress. In other words, smaller particles should be used to reduce intercalation-induced stresses.

As mentioned earlier, the model used here to simulate the intercalation-induced stress is a diffusion-stress coupling model. The effect of stress on diffusion will be discussed briefly using the 1D equations for a spherical particle. Substituting Eq. 25 into Eq. 13, we obtain

$$\mathbf{J} = -D(1 + 0c) \frac{\partial c}{\partial r} \quad [37]$$

In Eq. 37, $0c$ is always a positive number, and the effective diffusion coefficient is essentially $D(1 + 0c) > D$. Therefore, the diffusion is enhanced due to the extra term $0c$, which basically comes from the hydrostatic stress gradient term in Eq. 13. In other words, stress enhances the diffusion. This stress enhancement effect is also demonstrated numerically, as shown in Fig. 3. It shows the concentration profile at $t = 1000$ s with discharge current density $i = 2 \text{ A/m}^2$ on the surface. The profile including the effect of stress has a smaller gradient than that excluding the stress effect, confirming that stress enhances diffusion.

Substituting the material properties into the expression of $0 = 2K^2E/[9(1-\nu)RT]$, we obtain $0 = 1.557 \times 10^{-5} \text{ m}^3/\text{mol}$. If the maximum concentration is used, $0c_{\max} = 0.356$, which is not negli-

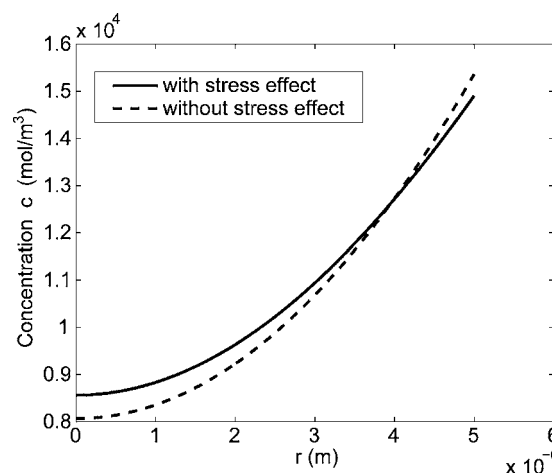


Figure 3. Numerical results for the effects of stress.

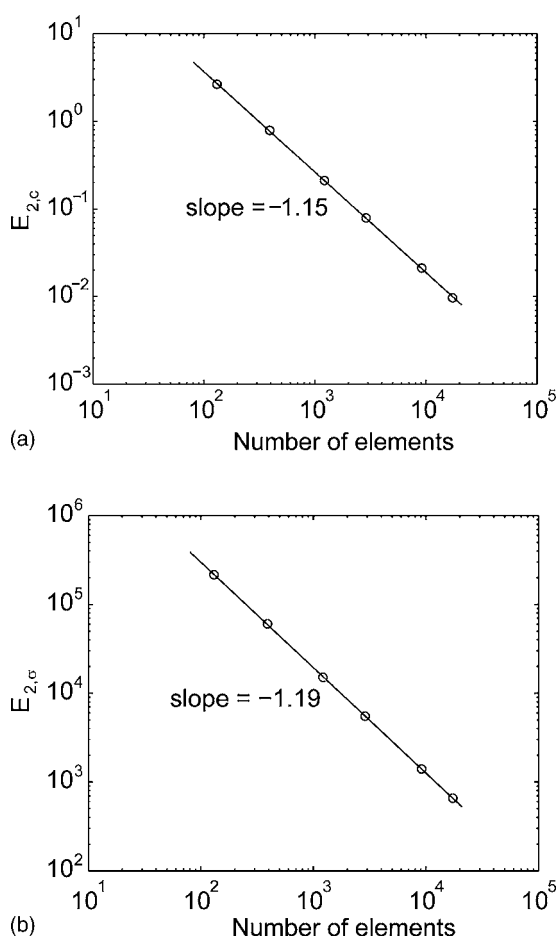


Figure 4. Convergence plot of finite element solutions for (a) hydrostatic stress and (b) concentration.

gible compared to unity. Therefore, stress effect cannot be neglected here for the case of LiMn_2O_4 . From the expression for σ , it can be observed that σ has smaller magnitude when the material has smaller modulus E and smaller partial molar volume K . Thus, the stress effect on diffusion may be negligible when the material is soft (i.e., having a low modulus).

3D finite element simulation results.— The 1D finite difference simulation, with 4001 grid points and a time step of 0.001 s, was used as the reference solution to study the **convergence of the finite element method**. Figure 4 shows the 2-norm errors (differences) between the finite element solutions and finite difference reference solutions at $t = 1000$ s. The parameters used in the simulations are current density $i = 2 \text{ A/m}^2$, and particle radius $r_0 = 5 \text{ }\mu\text{m}$. The finite element solutions converged to the reference solution as the number of elements used increased. At the same time, Fig. 4 also shows that solutions from 1D finite difference method and 3D finite element method were consistent, because the nondimensionalized errors of concentration and stress from 17,359 elements simulation were 6.5×10^{-7} and 1.5×10^{-5} , respectively (if nondimensionalized by the maximum values at $t = 1000$ s inside the particle).

To study the effect of aspect ratios on the intercalation-induced stress, ellipsoids with different aspect ratios were studied. The current density on the surface is fixed at $i = 2 \text{ A/m}^2$. For the ellipsoid, the lengths of three semi-axes a , b , and c satisfy $a = b$, and the aspect ratio is defined as $\alpha = c/a$, as sketched in Fig. 5. The volumes of the ellipsoids were fixed at $V = 4\pi \times 5^3/3 \text{ }\mu\text{m}^3$. A set of simulations, with different aspect ratios, was run by FEMLAB.

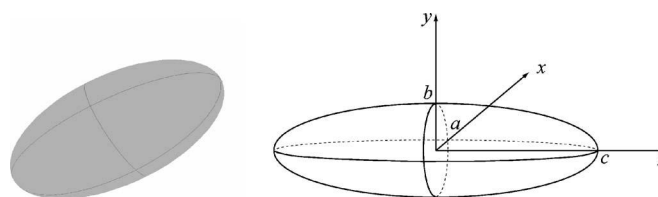


Figure 5. Schematic of an ellipsoidal particle, with coordinate system.

Characteristic solution profiles of concentration, von Mises stress and shear stress σ_{yz} , are shown in Fig. 6 at the end of the discharge process (when the surface concentration reaches the stoichiometric maximum) for an ellipsoid with aspect ratio 1.953. Figure 6 shows that (i) the concentration is higher around the poles, (ii) the von Mises stress is larger around the equator, and (iii) shear stress has its maximum on the surface. The solution profiles have the same patterns for other ellipsoids with different aspect ratios.

Figure 7 shows how the maximum von Mises stress inside the particle varies during the discharge process, for particles with different aspect ratios. It takes less time for particles with larger aspect ratios to completely discharge. Also, during discharge, von Mises stress increases first, and then drops. In Fig. 7, it can be observed that when aspect ratio increases, the stress increases first (for aspect ratios from 1.0 to 1.37) and then decreases (for aspect ratios from 1.37 to 3.81). For ellipsoids with aspect ratio 2.92 and 3.81, the intercalation-induced stress is smaller than that inside a sphere (aspect ratio 1.0).

Figure 8 shows how aspect ratio affects (a) peak value of maximum von Mises stress, and (b) peak value of maximum shear when the volumes of particles are fixed. Figure 8a shows that peak value of maximum von Mises stress inside the particle increases first and then decreases, as aspect ratio increases. For aspect ratios larger than 2.2, maximum von Mises stress decreases to less than that inside spherical particles (aspect ratio 1). Figure 8b shows that peak value of maximum shear stress decreases as aspect ratio increases. The results of Fig. 8 show that larger aspect ratios reduce the intercalation-induced stresses, over particles of lower aspect ratio when volume is preserved.

The peak values of maximum von Mises stresses are shown in Fig. 8a. Maximum stress first increases, then decreases with aspect ratio. This is due to two competing effects. When particle volume is preserved, increased aspect ratios result in increase of the longer semi-axis c , and reduction of shorter semi-axes a and b . Elongation of the longer semi-axis tends to increase maximum stress, while reduction of the shorter semi-axes tends to decrease the maximum stress. This competition results in a global maximum of stress at an aspect ratio of ~ 1.37 .

To further illustrate the effect of semi-axes on maximum stress, an additional set of simulations was performed in which the shorter semi-axes a and b were fixed, and aspect ratio α was increased by elongation of the longer semi-axis, c . Results obtained with a discharge current density $i = 2 \text{ A/m}^2$ are shown in Fig. 9. Stress first increases with aspect ratio because of the increase of the longer semi-axis, and then decreases slightly until asymptotically approaching the cylinder-fiber limit, i.e., $\alpha \rightarrow \infty$. As represented by the dashed line in Fig. 9, no physically relevant solutions are obtained for $\alpha > 7.9$, because the discharge process stops when the concentration on the particle surface reaches stoichiometric maximum, before the maximum stress actually reaches the peak value. To quantitatively illustrate this point, for an ellipsoid with aspect ratio 10, the stress reaches its peak value at $t = 721$ s. However, the simulation should terminate at $t = 617.5$ s, when the surface concentration already reaches stoichiometric maximum. The maximum stress at $t = 721$ s is 52.47 MPa, and the stress at $t = 617.5$ s is 52.26 MPa. Therefore, the stress when the process is terminated is only slightly smaller than the peak value.

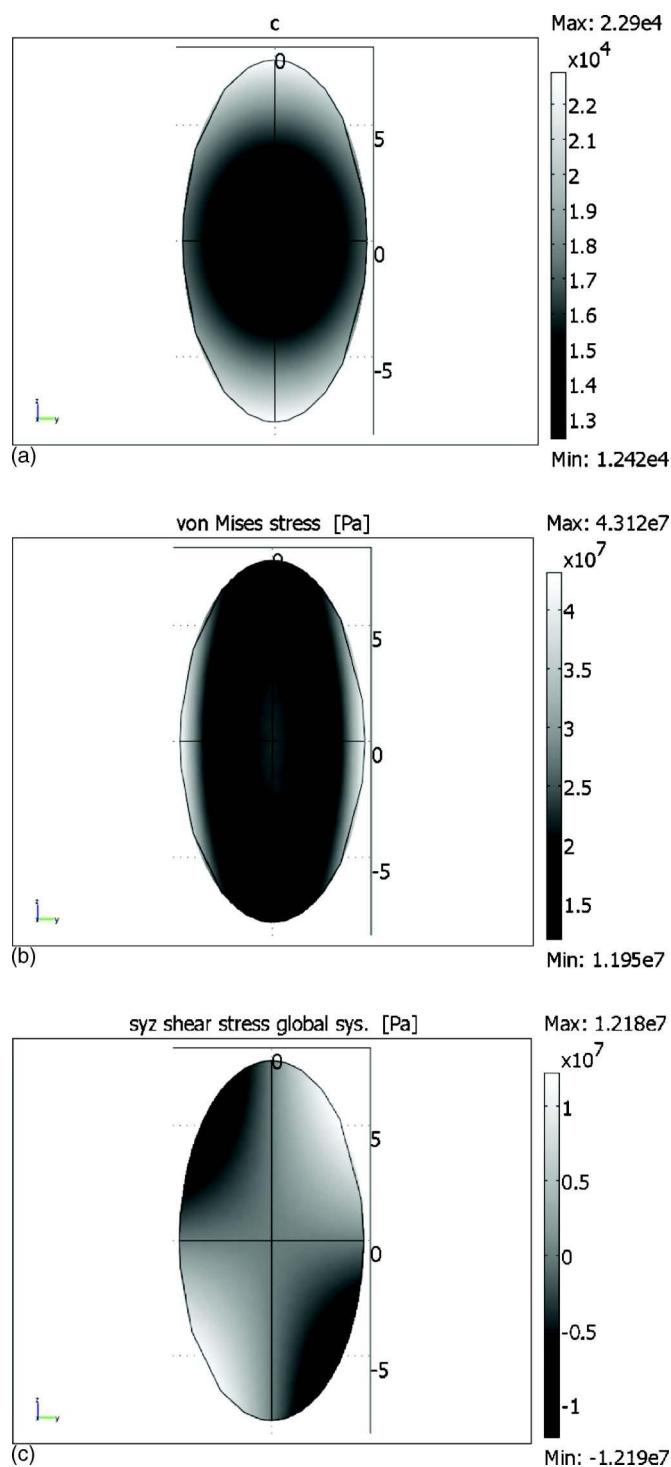


Figure 6. Solutions at the end of discharge for an ellipsoid of aspect ratio 1.953. (a) Concentration, (b) von Mises stress, and (c) shear stress σ_{yz} . (Dimensions in μm .)

Conclusion

Intercalation-induced stresses during the discharge process were simulated in this study using a stress-diffusion coupling model. Intercalation-induced stresses were simulated by analogy to thermal stress. Simulations of spherical particles show that larger particle sizes and larger discharge current densities give larger intercalation-induced stresses. Furthermore, internal stress gradients significantly enhance diffusion. Simulation results for ellipsoidal particles show

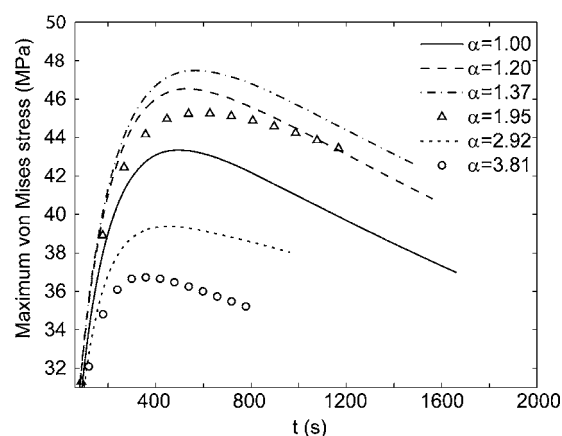


Figure 7. Maximum von Mises stress during discharge, for various ellipsoids.

that large aspect ratios are preferred to reduce the intercalation-induced stresses. In total, these results suggest that it is desirable to synthesize electrode particles with smaller sizes and larger aspect ratios to reduce intercalation-induced stress during cycling of lithium-ion batteries.

Acknowledgments

The authors gratefully acknowledge the support of our sponsors, including the U.S. Department of Energy through the BATT program (Dr. Tien Duong, Program Manager), the Army Research Of-

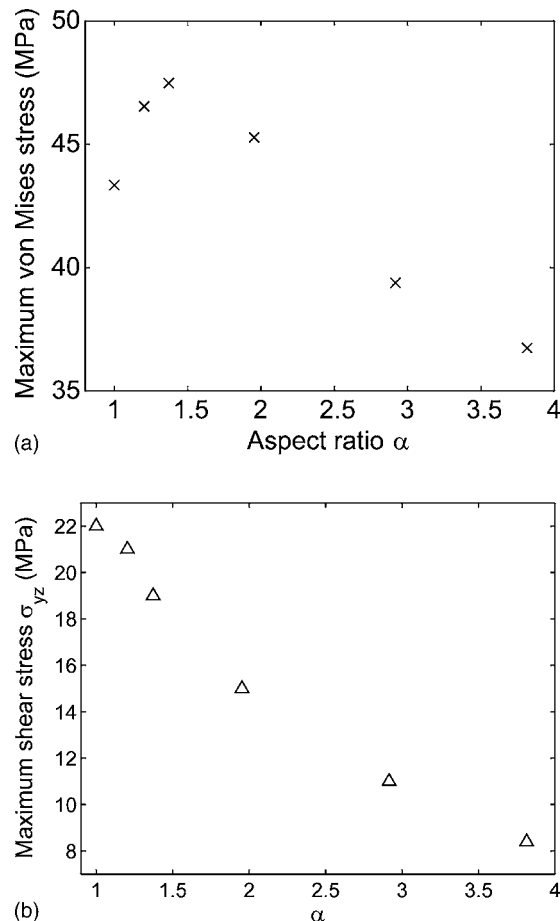


Figure 8. The effect of aspect ratio, for fixed particle volume.

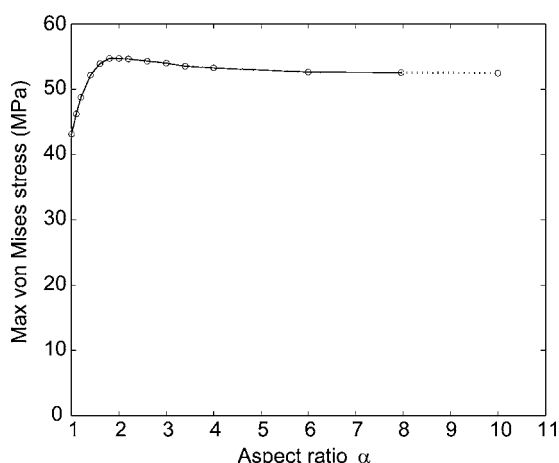


Figure 9. The effect of aspect ratio, for fixed shorter semi-axes.

fice (Dr. Bruce LaMattina, Program Manager), and the Ford Motor Company (Ted Miller and Kent Snyder, Program Managers). The authors also thank Professor James R. Barber and Dr. Chia-Wei Wang, both of the Department of Mechanical Engineering, University of Michigan, Ann Arbor, for helpful discussion and remarks.

University of Michigan assisted in meeting the publication costs of this article.

List of Symbols

a, b, c lengths of the three semi-axes of ellipsoid, m
 c concentration of lithium ions, mol/m³
 $\bar{c}$ concentration change from initial value, mol/m³
 D lithium diffusion coefficient, m²/s
 E Young's modulus, GPa
 F Faraday's constant, 96,487 C/mol
 I dimensionless current density
 i_n current density, A/m²
 J species flux, mol/(m² s)
 M mobility, m² mol/(J s)
 R gas constant, 8.314 J/(mol K)
 r_0 particle radius, μ m
 T temperature, K
 u displacement, m
 v ion movement velocity inside solid particles, m/s
 X molar fraction of lithium in the electrode

Greek

α aspect ratio
 ϵ_{ij} strain
 μ chemical potential, J/mol
 ν Poisson's ratio
 σ_{ij} stress, Pa
 V partial molar volume, m³/mol

Subscript

h hydrostatic (stress)
 $\max$ maximum
 r radial direction
 t tangential direction

Others

$\wedge$ dimensionless variables

References

- H. Wang, Y.-I. Jang, B. Huang, D. R. Sadoway, and Y.-M. Chiang, *J. Electrochem. Soc.*, **146**, 473 (1999).
- M.-R. Lim, W.-I. Cho, and K.-B. Kim, *J. Power Sources*, **144**, 3496 (1997).
- D. Wang, X. Wu, Z. Wang, and L. Chen, *J. Power Sources*, **140**, 125 (2005).
- W. I. F. David, M. M. Thackeray, L. A. de Picciotto, and J. B. Goodenough, *J. Solid State Chem.*, **67**, 316 (1987).
- J. Christensen and J. Newman, *J. Electrochem. Soc.*, **153**, A1019 (2006).
- Y.-B. Yi, C.-W. Wang, and A. M. Sastry, *J. Eng. Mater. Technol.*, **128**, 73 (2006).
- S.-I. Pyun, J.-Y. Go, and T.-S. Jang, *Electrochim. Acta*, **49**, 4477 (2004).
- Y.-H. Kim, S.-I. Pyun, and J.-Y. Go, *Electrochim. Acta*, **51**, 441 (2005).
- A. K. Padhi, K. S. Nanjundaswamy, and J. B. Goodenough, *J. Electrochem. Soc.*, **144**, 1188 (1997).
- J. Christensen and J. Newman, *J. Solid State Electrochem.*, **10**, 293 (2006).
- S. Prussin, *J. Appl. Phys.*, **32**, 1876 (1961).
- J. C. M. Li, *Metall. Trans. A*, **9A**, 1353 (1978).
- F. Yang, *Mater. Sci. Eng., A*, **409**, 153 (2005).
- M. Doyle, J. Newman, A. S. Gozdz, C. N. Schmutz, and C. M. Tarascon, *J. Electrochem. Soc.*, **143**, 1890 (1996).
- R. Darling and J. Newman, *J. Electrochem. Soc.*, **144**, 4201 (1997).
- P. Arora, M. Doyle, A. S. Gozdz, R. E. White, and J. Newman, *J. Power Sources*, **88**, 219 (2000).
- E. Deiss, D. Haringer, P. Novak, and O. Haas, *Electrochim. Acta*, **46**, 4185 (2001).
- R. Darling and J. Newman, *J. Electrochem. Soc.*, **145**, 990 (1998).
- C. Y. Ouyang, S. Q. Shi, Z. X. Wang, H. Li, X. J. Huang, and L. Q. Chen, *Europhys. Lett.*, **67**, 28 (2004).
- B. Ammundsen, J. Roziere, and M. S. Islam, *J. Phys. Chem. B*, **101**, 8156 (1997).
- S. P. Timoshenko and J. N. Goodier, *Theory of Elasticity*, McGraw-Hill, New York (1970).
- N. Noda, R. B. Hetnarski, and Y. Tanigawa, *Thermal Stresses*, 2nd ed., Taylor & Francis, New York (2003).
- W. R. McKinnon and R. R. Haering, in *Modern Aspects of Electrochemistry*, No. 15, R. E. White, J. O'M. Bockris, and B. E. Conway, Editors, Plenum Press, New York (1983).
- W. L. Wang, S. Lee, and J. R. Chen, *J. Appl. Phys.*, **91**, 9584 (2002).
- A. Paolone, R. Cantelli, G. Rousse, and C. Masquelier, *J. Phys.: Condens. Matter*, **15**, 457 (2003).